

LAB 4: CPU SCHEDULING ALGORITHMS

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1. FIRST COME FIRST SERVE (FCFS)

C PROGRAM:

```
#include<stdio.h>
int main()
{
    int n,i;
    int bt[20],wt[20],tat[20];
    float avg_wt=0,avg_tat=0;
    printf("Enter Number of Processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("Enter Burst Time for P%d: ",i+1);
        scanf("%d",&bt[i]);
    }
    wt[0]=0;
    for(i=1;i<n;i++)
        wt[i]=wt[i-1]+bt[i-1];
    for(i=0;i<n;i++){
        tat[i]=wt[i]+bt[i];
        avg_wt+=wt[i];
        avg_tat+=tat[i];
    }
    printf("\nProcess\tBT\tWT\tTAT\n");
    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);
    printf("\nAverage Waiting Time = %.2f", avg_wt/n);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat/n);
    return 0;
}
```

OUTPUT:

```
(kali@Harish)-[~]
$ nano fcfs.sh
(kali@Harish)-[~]
$ chmod +x fcfs.sh
(kali@Harish)-[~]
$ bash fcfs.sh
Enter Number of Processes
2
Enter Burst Time for P1
3
Enter Burst Time for P2
2
Process BT WT TAT
P1 3 0 3
P2 2 3 5
```

```
Average Waiting Time = 1.50
Average Turnaround Time = 4.00
```

2. SHORTEST JOB FIRST (SJF)

OUTPUT:

```
(kali@Harish)-[~]
$ nano sjf.sh
(kali@Harish)-[~]
$ chmod +x sjf.sh
(kali@Harish)-[~]
$ bash sjf.sh
Enter Number of Processes
2
Enter Burst Time for P1
1
Enter Burst Time for P2
2
Process BT WT TAT
P1 1 0 1
P2 2 1 3
Average Waiting Time = 0.50
Average Turnaround Time = 2.00
```

3. PRIORITY SCHEDULING

OUTPUT:

```
(kali@Harish)-[~]
$ nano priority.sh
(kali@Harish)-[~]
$ chmod +x priority.sh
(kali@Harish)-[~]
$ bash priority.sh
Enter Number of Processes
2
Enter Burst Time for P1
1
Enter Priority for P1
2
Enter Burst Time for P2
1
Enter Priority for P2
2
Process Priority BT WT TAT
P1 2 1 0 1
P2 2 1 1 2
Average Waiting Time = 0.50
Average Turnaround Time = 1.50
```

4. ROUND ROBIN SCHEDULING

OUTPUT:

```
(kali@Harish)-[~]  
└─$ nano rr.sh  
(kali@Harish)-[~]  
└─$ chmod +x rr.sh  
(kali@Harish)-[~]  
└─$ bash rr.sh  
Enter Number of Processes  
2  
Enter Burst Time for P1  
2  
Enter Burst Time for P2  
3  
Enter Time Quantum  
2  
Process BT WT TAT  
P1 2 0 2  
P2 3 2 5  
Average Waiting Time = 1.00  
Average Turnaround Time = 3.50
```